Diffusion-Based Data Augmentation for Imbalanced Medical Image Classification

MSML612 Group Project, Final Report
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Abstract

We asked whether diffusion-generated images can improve a downstream classifier on the rare classes of HAM10000, a severely imbalanced dermoscopic benchmark. We trained a class conditional 37.1M-parameter pixel-space DDPM from scratch on a leak-free lesion-level split and generated 1,000 images per class. In this class-balanced run, a nearest-neighbor check in two perceptual spaces, calibrated on real-to-real distances with whole-lesion exclusion, flags 97% of generated dermatofibroma (the rarest class) images as near-copies of specific training images; the flags trace to 81 of the 84 df training images, indicating that the generator is using all available information and not overemphasizing a select few. Flag rates generally rise across training checkpoints as train-referenced KID falls. A size-matched held-out analysis is less dramatic than the unmatched comparison: for df, median LPIPS is 0.134 to an equal-sized training reference and 0.155 to validation, and 66.0% of synthetic images are closer to train versus 63.7% for the real-data baseline. Filtering out the near-copies does not improve any of the seven held-out KID point estimates, although subset variability is too wide to establish each change inferentially. Across three classifier seeds, classical augmentation raises mean rare-class F1 from 0.343 to 0.573; unfiltered synthetic, filtered synthetic, and duplicated real (a dataset created by duplicating real images to match the length of the filtered synthetic dataset) score 0.299, 0.290, and 0.289. These descriptive results provide no evidence of a downstream gain from either synthetic arm and do not establish equivalence between filtered synthetic and duplicated real data. Overall, we can not conclude that diffusion-based augmentation or our filtering method improve the classification of rare classes.

1. Problem Statement and Research Question

Clinical image datasets are strongly unbalanced, with certain conditions having thousands of examples while others do not even have a hundred after the train-test split. Due to rare conditions lacking presence in these datasets, classifiers underperform exactly where reliable detection matters most. This is the motivation for augmenting the data to aid in the detection of rare classes. Classical augmentation only perturbs existing pixels; generative augmentation promises new samples. Diffusion models are the state of the art in image generation and provide native class-conditional control, which motivates our original question: can diffusion-generated images meaningfully improve a rare-class classifier? The project's findings sharpened that question into the one this report answers:

> *When a diffusion model is trained from scratch on tens of images per class, are its "synthetic" rare-class samples new images at all, and does anything they add to a classifier survive controlling for trivial oversampling?*

2. Data

HAM10000 (Tschandl et al., 2018) contains 10,015 dermoscopic images in 7 classes, from 66.9% melanocytic nevi (nv) down to 1.1% dermatofibroma (df). Many lesions are photographed more than once (4,501 of 10,015 images share a lesion with another image), so a naive image-level split leaks near-duplicates across the train/test boundary: averaged over 200 random image-level splits, 34.9% of test images share a lesion with a training image, and the rate is worst on the minority classes (mel 62.1%, df 51.1%). We therefore split at the lesion level, stratified by class, 70/15/15, seed 612, keeping all images; an assertion verifies that no lesion crosses splits. Full details are in the interim report; the resulting counts:

Class	**Condition Name**	**Train**	**Val**	**Test**	**Total**
nv	melanocytic nevi	4,679	1,015	1,011	6,705
mel	melanoma	776	163	174	1,113
bkl	benign keratosis-like lesions	765	165	169	1,099
bcc	basal cell carcinoma	354	77	83	514
akiec	Actinic keratoses and intraepithelial carcinoma	227	55	45	327
vasc	vascular lesions	96	23	23	142
df	dermatofibroma	84	15	16	115
All		**6,981**	**1,513**	**1,521**	**10,015**

From this point on, conditions will be referred to by their class name. akiec, vasc, and df will be considered rare classes. df, with 84 training images from 51 distinct lesions, is the stress test the study turns on.

3. Generative Model

The generator is a class-conditional DDPM (Ho et al., 2020) built on the diffusers `UNet2DModel` (von Platen et al., 2022), trained from scratch in pixel space at 64x64: 37.1M parameters, channels 128/256/256/256 with self-attention at 16x16 and 8x8, a cosine noise schedule with T = 1000 (Nichol and Dhariwal, 2021), and classifier-free guidance (Ho and Salimans, 2021) with label dropout 0.1. One shared model serves all classes so the rare classes can reuse representations learned from the common ones, and training uses class-balanced sampling so they receive equal gradient exposure. As run: 100,000 steps of AdamW (Loshchilov and Hutter, 2019) at 1e-4, effective batch 64 (batch 32 with 2-step gradient accumulation, which avoids VRAM spill on our 12 GB GPU), EMA decay 0.9999, 13.0 hours on a single GPU. Sampling uses DDIM (Song et al., 2021) at 50 steps with guidance 2.0 and seed 612; we generated 1,000 images per class.

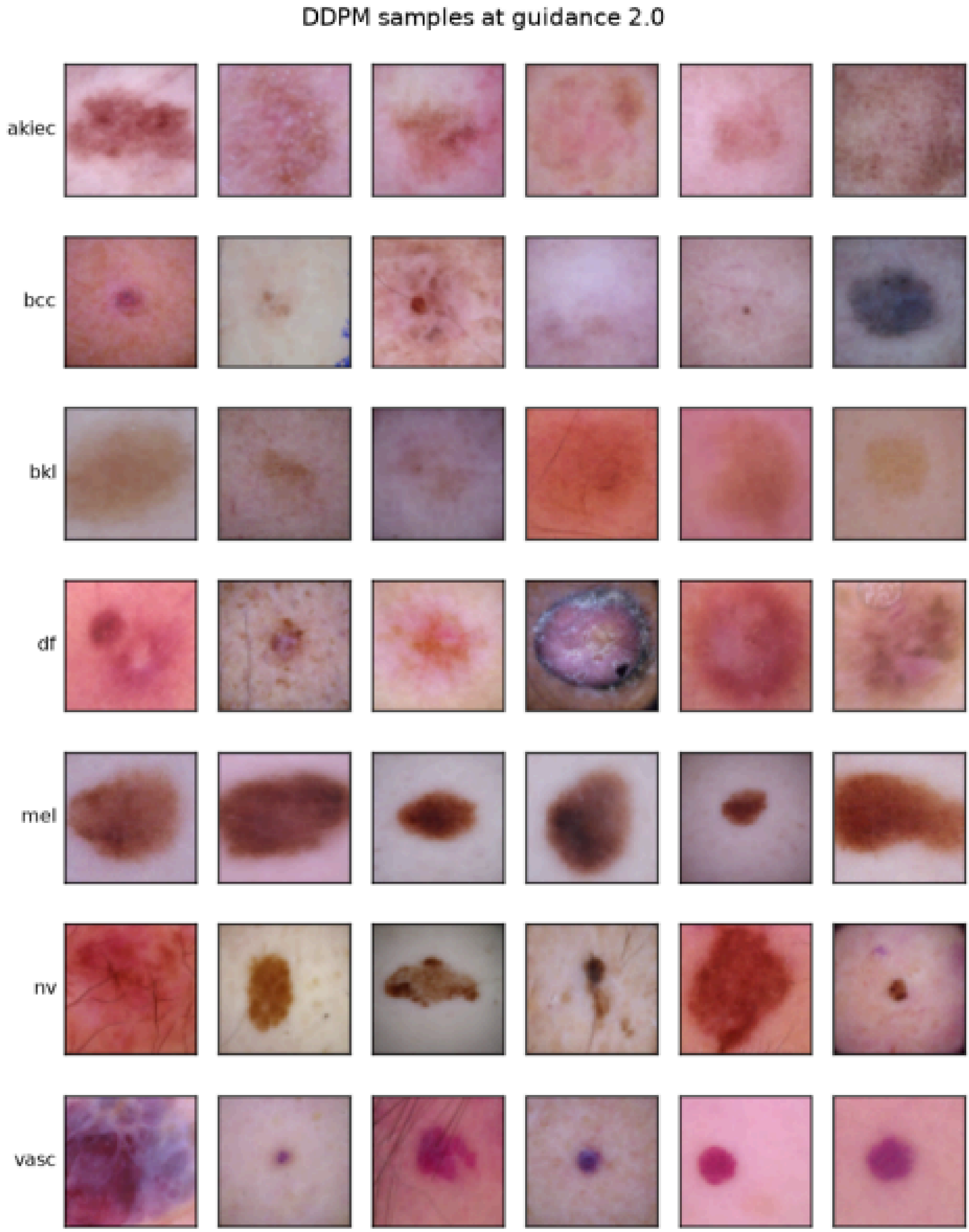

Figure 1. EMA samples at 100k steps, guidance 2.0. Legible lesion morphology in every class; the question the rest of the report answers is where that morphology comes from.

4. Generative Quality by the Standard Metrics

Per-class FID and KID against the real training images, with a finite-sample real-split reference formed by splitting each class's training set in half and scoring the halves. This reference is not a floor: it uses two half-sized real samples, whereas the synthetic comparison uses the full real class and 1,000 generated images, so the estimators have different finite-sample behavior. KID is reported x1000. For four classes (akiec, bcc, df, and vasc) the real-split FID reference exceeds the synthetic score, so FID is uninformative there and we rank those classes by KID only.

Class	FID	FID real-split ref.	KID	KID real-split ref.
akiec	46.52	70.96	14.10	-0.35
bcc	53.91	62.39	24.04	0.53
bkl	59.58	41.07	29.42	-0.15
df	38.33	107.42	5.68	3.75
mel	61.61	42.12	33.65	1.07
nv	52.30	11.14	24.56	-0.04
vasc	43.40	105.67	4.14	-5.27

The df and vasc reference entries also carry no variance: their halved train sets (42 and 48 images) sit below the 100-image KID subset cap, so every real-split subset is a permutation of the same images and the reported reference is a single draw, not a tight floor.

Taken at face value the two rarest classes, df and vasc, are the model's best. That reading is wrong, and the reason is the subject of Section 5: a generator that copies its training set can receive a low train-referenced KID by construction. These numbers are kept here because they are the metrics the field defaults to, and because the gap between this table and Section 5 is itself a finding: train referenced fidelity metrics cannot be read as generative quality without a memorization check next to them.

5. Memorization

5.1 Detection method

For every generated image we compute the nearest-neighbor distance to its class's real training images in two spaces: LPIPS (Zhang et al., 2018), computed through an exact algebraic decomposition that reduces it to squared Euclidean distance on precomputed vectors, and cosine distance on unit-normalized InceptionV3 pool features. An image is flagged when it falls below the 5th percentile of that class's real-to-real leave-one-out nearest-neighbor distribution in either space. Flagged images are quarantined, not deleted.

Calibration matters more than the metric. HAM10000's repeated lesion views mean a plain leave one-out null still contains sibling images of the same lesion, which drags the null distances down and makes the detector too lenient. We therefore recalibrated by excluding the entire lesion, not just the identical image, from each real image's neighbor set. This roughly doubles most thresholds (akiec LPIPS 0.033 to 0.092). All headline numbers below use the lesion-calibrated thresholds; the union of two 5% rules has a nominal false-positive rate between 5 and 10%, which is the baseline to read the table against.

5.2 The model copies its rare classes

Class	Flagged (image null)	Flagged (lesion null)	Surviving images
df	93.4%	97.0%	30
vasc	89.8%	92.3%	77
akiec	50.1%	79.1%	209
bcc	35.4%	56.2%	438
bkl	4.1%	25.0%	750
mel	4.6%	16.9%	831
nv	0.7%	2.8%	972

Flagged fraction is strongly associated with inverse class size. This is consistent with repeated exposure: under class-balanced sampling over 100k steps, each df training image was presented roughly 11,000 times. Class size and per-image exposure are coupled in this design, however, so this run does not identify the cause of the association. Visual inspection of the worst and near-threshold examples supports the copy interpretation and suggests that the threshold can miss visually similar pairs (Figure 2). Only nv, with 4,679 training images, stays below the nominal false-positive band.

Figure 2. The lowest-distance flagged pairs: each column pairs a generated image with its nearest real training image. These are near-pixel copies.

The copying is broad, not collapse onto a few prototypes: df's flags trace back to 81 of its 84 training images (50 of 51 lesions), vasc's to 88 of 96, akiec's to 163 of 227, and the most-copied df source accounts for only 8% of flags. Thus the detected copying spans most of the df and vasc training sets rather than a few sources.

5.3 Copying rises as train-referenced KID falls

We scored checkpoints from 10k to 100k steps, plus a guidance 1.0 probe at 100k (Figure 3). Flag rates generally rise with training while train-referenced KID falls. But these two curves are not independent: copies are close to the empirical training distribution by definition, so train-KID improves as copying worsens. The trajectory is also not perfectly monotonic at the point level (the 70k checkpoint beats 90k on both axes, and guidance 1.0 cuts df's flag rate from 97% to 84% at similar KID), so sampling choices move the numbers. What no tested setting produced is an operating point that is simultaneously novel and faithful: early checkpoints are novel but visibly unconverged (KID 52 to 106), late ones are faithful by copying.

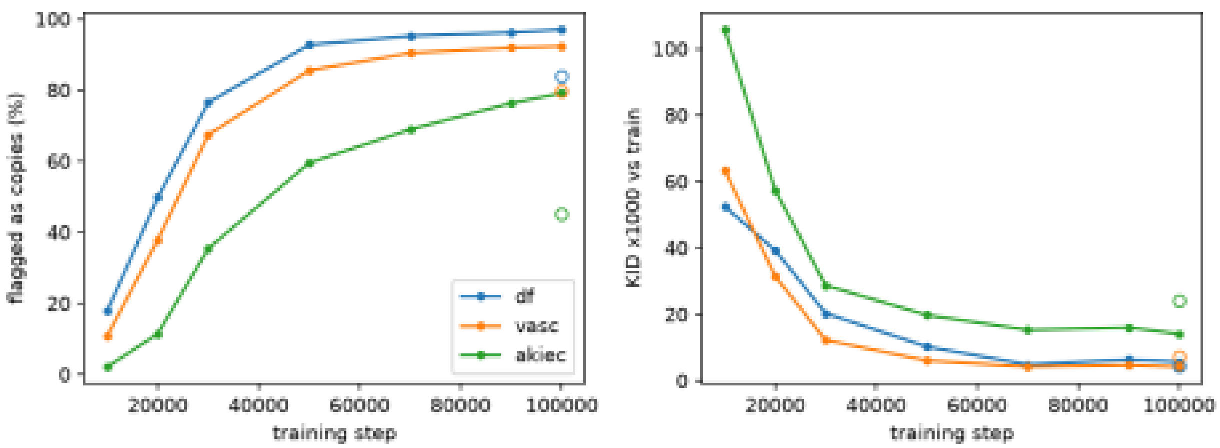

Figure 3. Left: lesion-calibrated flag rate. Right: KID x1000 vs train. Open markers: guidance 1.0 at 100k. Copying and train-referenced "quality" improve together, which is exactly why train referenced quality cannot be trusted here.

5.4 Held-out comparison with size-matched references

A nearest-neighbor comparison is sensitive to the number of candidate references. The training sets are larger than validation, so we repeatedly subsample each training class to the validation class size before comparing distances. The table reports median LPIPS and the fraction closer to the matched training reference, averaged over 20 seeded subsamples. For the real-data baseline, each validation image plays the role of a generated image; same-lesion validation neighbors are excluded.

Class	Pool to train, matched	Pool to val	Pool closer to train	Real val to train, matched	Real val to val	Real closer to train
df	0.134	0.155	66.0%	0.149	0.160	63.7%
vasc	0.112	0.153	73.1%	0.155	0.173	45.0%
akiec	0.122	0.135	64.6%	0.141	0.145	42.8%
bcc	0.137	0.145	61.4%	0.141	0.143	52.2%
bkl	0.145	0.148	55.8%	0.145	0.148	51.6%
mel	0.151	0.160	62.8%	0.146	0.149	57.6%
nv	0.125	0.125	52.1%	0.096	0.096	48.5%

Size matching changes the interpretation. For df, the train-to-validation distance ratio is 1.15 rather than 20.2, and the closer-to-train fraction is only 2.3 percentage points above the real baseline. For vasc the ratio is 1.37 and the fraction is 28.1 points above baseline; akiec also shows a substantial fraction difference. These held-out results are descriptive and mixed, not an independent twenty-fold copying result. The lesion-calibrated flag rates, source coverage, and inspected pairs in Section 5.2 remain the direct evidence of copying.

5.5 Filtering does not improve the held-out KID point estimates

The obvious salvage is to keep only the unflagged images. Measured against validation, all seven accepted-pool KID point estimates are higher than their unfiltered counterparts. The table reports KID x1000 mean +/- subset standard deviation, with real train scored against validation as a reference:

Class	Real train	All synthetic	Accepted only	Size-matched near-dup rate (all / accepted)
akiec	1.5 +/- 1.7	24.0 +/- 5.8	59.4 +/- 7.4	80% / 78%
bcc	4.1 +/- 2.7	43.6 +/- 7.8	69.9 +/- 7.5	72% / 68%
bkl	0.9 +/- 1.9	36.6 +/- 4.0	38.8 +/- 3.6	60% / 54%
df	6.7 +/- 9.0	15.0 +/- 14.6	26.1 +/- 10.3	76% / 30%
mel	1.3 +/- 1.8	40.5 +/- 5.4	43.0 +/- 5.2	49% / 44%
nv	-0.1 +/- 1.9	26.6 +/- 5.5	28.3 +/- 5.7	3% / 2%
vasc	8.7 +/- 6.5	21.1 +/- 7.6	33.7 +/- 9.0	73% / 62%

The held-out means support two observations. First, df's KID is 15.0 against validation rather than 5.7 against train, consistent with train-referenced KID rewarding copies. Second, filtering does not reveal a better-fidelity subset in these point estimates. The subset standard deviations are neither paired confidence intervals nor hypothesis tests; for example, df changes by 11.1 while the two subset standard deviations are 14.6 and 10.3. We therefore do not claim that filtering conclusively worsens every class.

The final column substantially reduces the original pool-size confound. For each of 20 repeats, the synthetic pool and real calibration set are subsampled to the same size: the smaller of the pool size, class training size, and 500. The LPIPS threshold is recomputed on that real subset with whole-lesion exclusion before the synthetic subset is scored. The matched real null rate is 4.8 to 5.3% except for accepted df, whose 30-image quantile yields 6.7%. The rare-class pool rates therefore remain far above their matched nulls after matching the number of images evaluated. They are detector rates under this LPIPS rule, not estimates of duplicate prevalence.

6. Downstream Comparison

The generative analysis motivates comparing synthetic data with a duplicated-real control. The classifier architecture follows the interim plan: a ResNet-18 (He et al., 2016) trained from scratch at 64x64 with class-balanced sampling. Architecture, AdamW optimizer, learning rate schedule, 6,000-step budget, evaluation cadence, and seed IDs are fixed across regimes. The arm manifest or train time augmentation is the regime-defining difference. Five regimes were run for three seeds each (612, 613, 614):

1. Real data only.
2. Real + classical augmentation (flips, rotation, color jitter).
3. Real + duplicated-real oversampling (matching regime 5's added volume with copies of real images: the control the memorization result makes essential).
4. Real + unfiltered synthetic (1,000 per class).
5. Real + filtered synthetic (the accepted pools of Section 5.5).

The real-only and classical arms use the 6,981-row real manifest, duplicated-real and filtered synthetic each add 3,307 class-matched rows, and unfiltered synthetic adds 7,000. These are fixed manifests, not validation-selected mixing ratios. Validation macro-F1 selects the checkpoint only; its test set is then evaluated once for that seed and regime. The primary summary is mean F1 on akiec, df, and vasc, with macro-F1 and balanced accuracy secondary. Results below are mean +/- sample standard deviation across three seeds.

Regime	**Rare-class F1 (delta vs real)**	**Macro F1**	**Balanced acc.**
Real only	0.343 +/- 0.016 (reference)	0.438 +/- 0.014	0.422 +/- 0.027
Classical aug	0.573 +/- 0.040 (+0.231)	0.604 +/- 0.025	0.611 +/- 0.018
Duplicated-real	0.289 +/- 0.011 (-0.054)	0.420 +/- 0.002	0.446 +/- 0.021
Unfiltered synthetic	0.299 +/- 0.022 (-0.044)	0.432 +/- 0.012	0.414 +/- 0.018
Filtered synthetic	0.290 +/- 0.048 (-0.052)	0.416 +/- 0.029	0.399 +/- 0.033

Regime	akiec F1	bcc F1	bkl F1	df F1	mel F1	nv F1	vasc F1
Real only	0.232	0.429	0.375	0.068	0.394	0.843	0.727
Classical aug	0.487	0.612	0.518	0.395	0.514	0.867	0.838
Duplicated-real	0.184	0.433	0.409	0.111	0.409	0.824	0.571
Unfiltered synthetic	0.181	0.496	0.401	0.060	0.385	0.846	0.656
Filtered synthetic	0.118	0.432	0.414	0.032	0.365	0.832	0.721

Classical augmentation has the highest three-seed mean for every rare class and raises mean rare class F1 by 0.231 over real only. Neither synthetic arm improves the corresponding mean. Filtered synthetic and duplicated real have nearly equal aggregate rare-class means, a difference of +0.0016, but this is cancellation rather than demonstrated equivalence: filtered-minus-duplicated differences are -0.066 for akiec, -0.080 for df, and +0.150 for vasc. Their paired rare-class differences by seed are +0.0248, -0.0637, and +0.0436. With only three seeds and no prediction-level bootstrap or predeclared equivalence margin, these numbers show no observed synthetic-data gain in this experiment, not that filtered synthetic and duplicated real are interchangeable in the population. Per run values are in `reports/classifier_runs.csv`, with arm summaries in `reports/classifier_summary.csv`.

Only vasc shows a large improvement in F1 score from duplicated-real to filtered synthetic. While vasc is a rare class, it has the second highest F1 score under every regime, implying the classifier did not particularly struggle to identify it. The highest scoring class across all regimes, nv, does not show much change in F1 score across regimes. The two classes that score lowest in the duplicated-real regime, akiec and df, achieve worse results under the filtered synthetic regime. Both of these are rare classes. Overall it can not be concluded that diffusion-based augmentation is helpful for the classification of rare classes.

Furthermore, filtered synthetic does not show significant improvement over unfiltered synthetic. vasc and bkl are the only classes to increase their F1 scores after the filter is applied. bkl's F1 score only increases by .013, so caution should be taken before asserting that the class benefits from the filter. However, the rare classes with low F1 scores, akiec and df, achieve slightly worse results after filtration (-0.063 and -0.028 respectively). Therefore, there is no reason whatsoever to conclude that removing copies improves classification for these classes.

When evaluating our synthetic datasets vs the duplicated-real one, it can be seen from the first table that the rare-class F1 scores are within one standard deviation of each other. These calculations include vasc, which is an outlier among rare classes as the only one with high F1 scores. vasc's F1 score improves monotonically from duplicated-real to unfiltered synthetic to filtered synthetic as our experiment predicted, but the other rare classes decrease monotonically in the same direction. Therefore, the use of diffusion to compensate for poor classification in rare classes might be even worse than the results show. It is also worth noting that all three regimes have predicted F1 scores worse than with no augmentation.

In addition, classical augmentation achieves the best F1 scores for every class. The lowest scoring classes across all regimes, akiec and df, have the biggest improvements under classical augmentation relative to the other regimes (+0.255 and +0.327 compared to reference values, compared to +0.183, +0.143, +0.120, +0.024, and +0.111 for bcc, bkl, mel, nv, and vasc respectively).

7. Discussion

Plausible Copying Mechanism: Class-balanced sampling, our deliberate defense against rare-class neglect, multiplied each df image's expected exposure by roughly 12x relative to natural frequency, to about 11,000 presentations over the run. This makes repeated exposure a plausible contributor to the inverse class-size pattern. Exposure and class size are coupled, however, and there is no natural frequency control, so this study cannot identify balancing as the cause. The supported scope is narrower: in this one 37M-parameter from-scratch DDPM run, the smallest classes have the highest calibrated detection rates.

Duplicated-Real vs Filtered Synthetic: This is the main comparison our paper focuses on. Removing near-copies from the synthetic dataset ensures that the augmentation is only images new to the classifier. Duplicating real images eliminates any differences caused by dataset size without adding any new information. Therefore, this comparison examines solely if the generator can contribute meaningful information to the classifier. Based on our results, we can not conclude that this is the case. The average F1 scores for rare classes are near identical, and filtered synthetic augmentation might have a negative effect on classes with poor classification. A possible explanation for this is that with a small training dataset, the diffusion model can not learn abstract features. As a result, the high quality output for rare classes is mostly near copies. This explanation is consistent with rare classes having high flagging rates for duplicates. When these duplicates are filtered out, much of the remaining synthetic data does not accurately represent the class. Adding false information is not better than adding no information, therefore our experimental augmentation does not help classify rare classes.

Filtered Synthetic vs Unfiltered Synthetic: Comparing the classifiers trained on the synthetic dataset before and after near-copies are removed isolates for the effect of our deduping process. We fail to demonstrate that this process improves classification, and even have some evidence to the contrary. To continue with the explanation from the previous paragraph, removing near-copies increases the proportion of data taken up by inaccurate images. The differences between these regimes are small, however, so it is possible that the classifier architecture is resilient to this. Another explanation for the similarity in results is that both regimes have drawbacks that somewhat cancel out. For example, the unfiltered synthetic dataset has significantly more images than the filtered dataset. Possible scenarios could be: the unfiltered dataset has higher quality data but suffers from overfitting, or maybe the filtering process does actually increase the density of useful information in the dataset ,but is held back by having less images to train on.

Real vs Synthetic: Our two synthetic datasets perform very similarly on the rare classes as simply duplicating real data to match the size of the filtered synthetic dataset. In fact, the only class that contributes to the difference in favor of the synthetic datasets does not have poor classification at all. Arguably, every class that this experiment aims to help performs worse on the synthetic datasets. The intuitive reason for this is that if the diffusion model could learn the necessary features to distinguish a class using only the real images, the dataset already contains all the information the classifier needs. However, this is not necessarily true, as the two models have different internal representations. This means experimentation is necessary to validate this intuition. A clear example of this is that classical augmentation, which does not generate original information, significantly improves classification. This shows that training generative models on the same dataset as the classifier is not inherently pointless and should remain a topic of study.

Recommendation: Based on our results we can not recommend diffusion models as a way to augment class-imbalanced data. We are unable to produce results supporting improved classification over the unaugmented dataset. Given the significant computational cost of training and running a generative model, definitive gains would be necessary to justify it. Of all the augmentation methods tested, classical augmentation is the clear winner. With the best F1 scores and low cost, we recommend continuing to use classical augmentation instead of switching to diffusion.

Why We Did Not Retrain: The candidate mitigations, in the order we would try them, are capping the oversampling weight (e.g. 90% empirical + 10% balanced sampling, which cuts per-image exposure by 5 to 10x), dihedral train-time augmentation with an augmentation-aware detector, a smaller model, scheduled guidance, and memorization-based early stopping. Each costs a full 13-hour training run plus re-scoring, and a single unreplicated attempt would support weaker claims than the analysis above. We chose to spend the remaining budget characterizing the negative result and measuring its downstream cost. The honest statement of scope: across the tested checkpoints, guidance scales, and sampler, no acceptable operating point was observed; the experiment does not estimate true copy prevalence (the detector's null rate is 5 to 10%, and it under-flags visually confirmed near-copies), identify a single cause, or rule out untested mitigations.

Limitations: Everything is established at 64x64 on one generator run. The validation references for the rarest classes are small (df 15, vasc 23 images), so held-out KID has substantial subset variability. Size matching also shows that the earlier unmatched train-versus-validation distance contrast was mostly a reference-count effect for df. Classifier results cover three seeds and have no inferential or equivalence analysis. Flag rates are detection rates under a calibrated threshold, not prevalence estimates. The ethical stake is worth stating plainly: in a medical setting, lightly perturbed copies of real patient images can create privacy risk while being presented as augmentation, so a memorization check should accompany such a pipeline. The run JSONs did not record immutable code and input hashes at execution time; the supplied run manifest is a post-hoc artifact inventory, not execution attestation.

8. Deviations from the Interim Plan

- **The StyleGAN2-ADA baseline was dropped.** Once the memorization result landed, the informative comparison became synthetic-vs-duplicated-real, not diffusion-vs-GAN. A GAN trained on the same data would require its own memorization and downstream evaluation, and the available compute went to the checkpoint sweep and held-out analysis instead. No GAN evidence was collected, so this report makes no diffusion-vs-GAN claim.
- **Four regimes became five.** The replication control the interim report mentioned in passing (duplicated-real oversampling) was promoted to the decisive comparator, and the

diffusion regime split into unfiltered and filtered variants.

- **Classifier replication and inference were reduced.** The interim protocol specified five paired seeds, per-class Wilcoxon tests with Holm correction, and prediction-level bootstrap intervals. Execution used three seeds and did not save prediction-level outputs, so none of those inferential analyses was performed. The report therefore treats the classifier results as descriptive means and standard deviations. Moreover, a two-sided exact signed-rank test with only five nonzero pairs could not attain $p < 0.05$, so the originally specified seed count was itself insufficient for that test.
- **Ratio selection and planned ablations were omitted.** The executed arms use one fixed manifest each; validation selects checkpoints, not mixing ratios. The 0/25/50/100% ratio sweep and the 25/50/100% generator sample-size study were not run.
- **The guidance sweep narrowed.** Instead of {1.0, 2.0, 3.0} for sample quality, we probed guidance 1.0 specifically for its effect on copying (Figure 3).

9. Related Work

The diffusion foundations are as in the interim report: DDPM (Ho et al., 2020), the cosine schedule (Nichol and Dhariwal, 2021), classifier-free guidance (Ho and Salimans, 2021), and the diffusers implementation (von Platen et al., 2022). Frid-Adar et al. (2018) and Trabucco et al. (2024) report generative-augmentation gains; our result is a caution about the regime where those gains are claimed with the least data. On measurement, FID is due to Heusel et al. (2017), KID to Binkowski et al. (2018), and LPIPS to Zhang et al. (2018). Our held-out comparison follows the train-versus-test distance intuition of Meehan et al. (2020), but it does not implement their formal three-sample statistic or hypothesis test. Carlini et al. (2023) extracted training images from large diffusion models; Dar et al. (2025) documented the same phenomenon in medical diffusion models, including copies that survive augmented training, which informed our choice not to treat augmentation alone as a guaranteed fix.

10. References

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